

Arrays Practice Assignment

Part - 2

Note: Create Array with User Input Using Scanner

Que 1. Write a program to swap two elements of an array at given indices i and j.

Example

Input: arr = [10, 20, 30, 40, 50], i = 1, j = 3

Output: Array after swapping: [10, 40, 30, 20, 50]

Explanation: The elements at indices 1 and 3 (20 and 40) are swapped, resulting in the new array [10, 40, 30, 20, 50]

Que 2. Write a Java program to reverse an array using the swapping technique.

Example

Input: arr = {10, 20, 30, 40, 50}

Output: 50 40 30 20 10

Explanation: Swap the first and last elements, moving inward until the middle, to reverse the array.

Que 3. Write a Java program to swap the portion of the array after the given index k. The portion of the array starting from index k+1 to the end should be reversed.

Example

Input: arr = {1, 2, 3, 8, 7, 5, 4} Index (k): 2

Output: 1 2 3 4 5 7 8

Explanation: The array after index k (portion {8, 7, 5, 4}) is reversed to {4, 5, 7, 8}. The array till index 2 remains unchanged.

Que 4. Write a program to return the second largest element from the array. If the second largest element doesn't exist then return -1.

Note: The second largest element should not be equal to the largest element.

Example

Input: arr[] = [12, 35, 1, 10, 34, 1]

Output: 34

Explanation: The largest element of the array is 35 and the second largest element is 34.

Input: arr[] = [10, 5, 10]

Output: 5

Explanation: The largest element of the array is 10 and the second largest element is 5.

Input: arr[] = [10, 10, 10]

Output: -1

Explanation: The largest element of the array is 10 and the second largest element does not exist.

Que 5. Write a program to return the second smallest element from the array. If the second smallest element doesn't exist, return -1.

Note: The second smallest element should not be equal to the smallest element.

Example

Input: arr[] = [12, 35, 1, 10, 34, 1]

Output: 10

Explanation: The smallest element in the array is 1, and the second smallest element is 10.

Input: arr[] = [10, 5, 10]

Output: 10

Explanation: The smallest element in the array is 5, and the second smallest element is 10.

Input: arr[] = [10, 10, 10]

Output: -1

Explanation: All elements are the same, so no second smallest element exists.

Que 6. Write a program to return the kth largest element from an array. If the array has less than k distinct elements, return -1.

Example

Input: arr[] = [3, 2, 1, 5, 6, 4], k = 4

Output: 3

Explanation: The distinct elements of the array are [6, 5, 4, 3, 2, 1] in sorted order. The 4th largest element is 3.

Input: arr[] = [10, 10, 20, 20], k = 3

Output: -1

Explanation: The distinct elements of the array are [10, 20]. There are less than 3 distinct elements, so the output is -1.

Que 7. Write a program to return the kth smallest element from an array. If the array has less than k distinct elements, return -1.

Example

Input: arr[] = [3, 3, 2, 1, 4, 4], k = 3

Output: 3

Explanation: The distinct elements of the array are [1, 2, 3, 4] in sorted order. The 3rd smallest element is 3.

Input: arr[] = [8, 8, 8, 8], k = 2

Output: -1

Explanation: All elements in the array are the same (8). Therefore, there is no 2nd smallest distinct element, as only one distinct element (8) exists.

Que 8. Write a program to check if an array of integers is sorted in non-decreasing order. If the array is sorted, print true; otherwise, print false.

Example

Input: arr[] = [1, 2, 3, 4, 5]

Output: true

Explanation: The array is sorted in non-decreasing order.

Input: arr[] = [5, 4, 3, 2, 1]

Output: false

Explanation: The array is sorted in decreasing order, not non-decreasing order.

Input: arr[] = [10, 20, 20, 30]

Output: true

Explanation: The array contains equal elements but is sorted in non-decreasing order.

Input: arr[] = [1, 3, 2, 4, 5]

Output: false

Explanation: The array is not sorted due to the unordered pair (3, 2).

Que 9. Write a program to count the number of unique elements in an array of integers, including duplicates.

Example:

Input: arr[] = [10, 20, 20, 30, 40, 40, 40]

Output: 4

Explanation: Unique elements are 10, 20, 30, 40.

Input: arr[] = [1, 1, 1, 1]

Output: 1

Explanation: Unique element is 1.

Que 10. Write a program to remove duplicates from an array and print the new array containing only unique elements.

Example:

Input: arr[] = [10, 20, 20, 30, 40, 40, 40]

Output: [10, 20, 30, 40]

Explanation: Duplicate elements 20 and 40 are removed.

Input: arr[] = [1, 1, 1, 1]

Output: [1]

Explanation: Only the unique element 1 remains.

Que 11. Write a program to rotate an array to the right by 1 position. The last element of the array should become the first, and all other elements should shift one position to the right.

Example:

Input: arr[] = [1, 2, 3, 4, 5]

Output: [5, 1, 2, 3, 4]

Explanation: The last element 5 becomes the first, and the rest shift to the right.

Input: arr[] = [10, 20, 30, 40]

Output: [40, 10, 20, 30]

Explanation: The last element 40 becomes the first, and others move right.

Input: arr[] = [7]

Output: [7]

Explanation: A single-element array remains the same after rotation.

Que 12. Write a program to rotate an array to the left by 1 position. The first element of the array should move to the last, and all other elements should shift one position to the left.

Example:

Input: arr[] = [1, 2, 3, 4, 5]

Output: [2, 3, 4, 5, 1]

Explanation: The first element 1 moves to the end, and the rest shift to the left.

Input: arr[] = [10, 20, 30, 40]

Output: [20, 30, 40, 10]

Explanation: The first element 10 moves to the end, and others move left.

Input: arr[] = [7]

Output: [7]

Explanation: A single-element array remains the same after rotation.

Que 13. Write a program to rotate an array by k positions. Each element in the array should be shifted to the left k times, and the first k elements should move to the end of the array.

Example:

Input: arr[] = [1, 2, 3, 4, 5], k = 2

Output: [3, 4, 5, 1, 2]

Explanation: The first two elements [1, 2] move to the end.

Input: arr[] = [10, 20, 30, 40, 50, 60], k = 4

Output: [50, 60, 10, 20, 30, 40]

Explanation: The first four elements [10, 20, 30, 40] move to the end.

Que 14. Given an integer array nums, move all 0s to the end of the array while maintaining the relative order of the non-zero elements. You must solve this problem in- place, without using extra space for another array.

Example:

Input: nums = [0, 1, 0, 3, 12]

Output: [1, 3, 12, 0, 0]

Explanation: The 0s are moved to the end of the array while the relative order of the non-zero elements remains the same.

Que 15. Write a program that takes two arrays as input and merges them into a single array.

The program should return the new merged array containing all elements from the first array followed by all elements from the second array.

Example

Input: arr1 = [10, 20, 30] arr2 = [40, 50, 60]

Output: Merged Array: [10, 20, 30, 40, 50, 60]

Explanation: First, add all elements from Array 1, then add all elements from Array 2 to the new array.

The resulting array [10, 20, 30, 40, 50, 60] is returned as the output.

Que 16. Given two sorted arrays a[] and b[], where each array may contain duplicate elements, the task is to return the array of elements in the union of the two arrays in sorted order. The union should only contain distinct elements, even if an element appears multiple times in the arrays.

Example:

Input:

a[] = [1, 2, 3, 4, 5]

b[] = [2, 4, 6, 8]

Output:

[1, 2, 3, 4, 5, 6, 8]

Explanation: The union of the two arrays is [1, 2, 3, 4, 5, 6, 8], as it includes only distinct elements.

Input:

a[] = [1, 2, 2, 4, 5]

b[] = [2, 4, 5, 5, 6]

Output:

[1, 2, 4, 5, 6]

Explanation: The union of the two arrays is [1, 2, 4, 5, 6], with duplicates removed.

Que 17. Given an array `nums[]` containing `n` distinct numbers in the range `[0, n]`, return the only number in the range that is missing from the array.

The array contains all numbers from 0 to `n` except one. Your task is to find the missing number efficiently.

Example:

Input: `nums = [1, 2, 4, 5, 0]`

Output: 3

Explanation: The number 3 is missing from the array, as all other numbers from 0 to 5 are present.

Input: `nums = [0, 2]`

Output: 1

Explanation: The number 1 is missing from the array.

Input: `nums = [9, 6, 4, 2, 3, 5, 7, 0, 1]`

Output: 8

Explanation: The number 8 is missing from the array.

Que 18. Given a binary array `nums`, return the maximum number of consecutive 0's in the array.

Example

Input:

`nums = [0, 0, 1, 0, 0, 0]`

Output: 3

Explanation: The last three digits are consecutive 0's. The maximum number of consecutive 0's is 3

Input: `nums = [1, 0, 0, 1, 0, 1]`

Output: 2

Explanation: The two digits(index 1 and 2) are consecutive 0's. The maximum number of consecutive 0's is 2.

Que 19. You are given array of integers, `nums`, where every element appears twice except for one. Your task is to return the element that appears only once.

Example

Input: `nums = [4, 1, 2, 1, 2]`

Output: 4

Explanation: The number 4 appears only once, while 1 and 2 each appear twice.

Que 20. Given an array arr. Find the majority element in the array. If no majority exists, return -1. A majority element in an array is an element that appears strictly more than $\text{arr.size()} / 2$ times in the array.

Example

Input : arr[] = [1, 1, 2, 1, 3, 5, 1]

Output : 1

Explanation: Note that 1 appear 4 times which is more than $7 / 2$ times

Input : arr[] = [3, 3, 4, 2, 4, 4, 2, 4]

Output : -1

Explanation: There is no element whose frequency is greater than the half of the size of the array size.

Input : arr[] = [3]

Output : 3

Explanation: Appears more than $n/2$ times

Que 21. Write a program to swap the contents of two arrays of the same length. After the swap, the contents of the first array should become those of the second array and vice versa.

Example

Input: arr1 = [1, 2, 3], arr2 = [4, 5, 6]

Output: After swapping:

arr1 = [4, 5, 6], arr2 = [1, 2, 3]

Explanation: The contents of array1[] become those of array2[], and the contents of array2[] become those of array1[].

Que 22. Write a program to find the peak element in an unsorted array.

Note:

- An element is a peak if it is greater than or equal to its neighbours.
- For the first element, it is a peak if it is greater than or equal to the second element.
- For the last element, it is a peak if it is greater than or equal to the second-to-last element.
- For other elements, it is a peak if it is greater than or equal to both its neighbours.

If there are multiple peak elements, return any one of them.

Example

Input: arr[] = {5, 10, 20, 15}

Output: 20

Explanation: The element 20 has neighbours 10 and 15, both of them are less than 20.

Input: arr[] = {10, 20, 15, 2, 23, 90, 90}

Output: 20 or 90

Explanation: The element 20 has neighbours 10 and 15, both of them are less than 20, similarly 90 has neighbours 23 and 67.

Que 23. Given an array `arr[]` of size `n`, the task is to find all the Leaders in the array. An element is a Leader if it is greater than all the elements to its right side.

Note: The rightmost element is always a leader.

Example

Input: `arr[] = [16, 17, 4, 3, 5, 2]`

Output: `[17 5 2]`

Explanation: 17 is greater than all the elements to its right i.e., [4, 3, 5, 2], therefore 17 is a leader. 5 is greater than all the elements to its right i.e., [2], therefore 5 is a leader. 2 has no element to its right, therefore 2 is a leader.

Input: `arr[] = [1, 2, 3, 4, 5, 2]`

Output: `[5 2]`

Explanation: 5 is greater than all the elements to its right i.e., [2], therefore 5 is a leader. 2 has no element to its right, therefore 2 is a leader.

Que 24. Given an array of integers `nums` and an integer `target`, return the indices of the two numbers such that they add up to the target.

You may assume that each input would have exactly one solution, and you may not use the same element twice.

You can return the answer in any order.

Example

Input: `nums = [2,7,11,15]`, `target = 9`

Output: `[0,1]`

Explanation: Because `nums[0] + nums[1] == 9`, we return `[0, 1]`.

Input: `nums = [3,2,4]`, `target = 6`

Output: `[1,2]`

Explanation: Because `nums[1] + nums[2] == 6`, we return `[1, 2]`.

Input: `nums = [3,3]`, `target = 6`

Output: `[0,1]`

Explanation: Because `nums[0] + nums[1] == 6`, we return `[0, 1]`.

Que 25. Given an array `arr[]` containing integers and an integer `k`, your task is to find the length of the longest subarray where the sum of its elements is equal to the given value `k`. It is guaranteed that a valid subarray exists.

Example

Input: `arr[] = [10, 5, 2, 7, 1, 9]`, `k = 15`

Output: 4

Explanation: The subarray `[5, 2, 7, 1]` has a sum of 15 and length 4.

Input: `arr[] = [-1, 2, -3]`, `k = -2`

Output: 3

Explanation: The subarray `[-1, 2, -3]` has a sum of -2 and length 3.

Input: `arr[] = [1, -1, 5, -2, 3]`, `k = 3`

Output: 4

Explanation: The subarray `[1, -1, 5, -2]` has a sum of 3 and a length 4.

Que 26. Given an array of integers of size N, the task is to find the first non-repeating element in this array.

Example

Input: [-1, 2, -1, 3, 0]

Output: 2

Explanation: The first number that does not repeat is:
2

Input: [9, 4, 9, 6, 7, 4]

Output: 6

Explanation: The first number that does not repeat is:
6